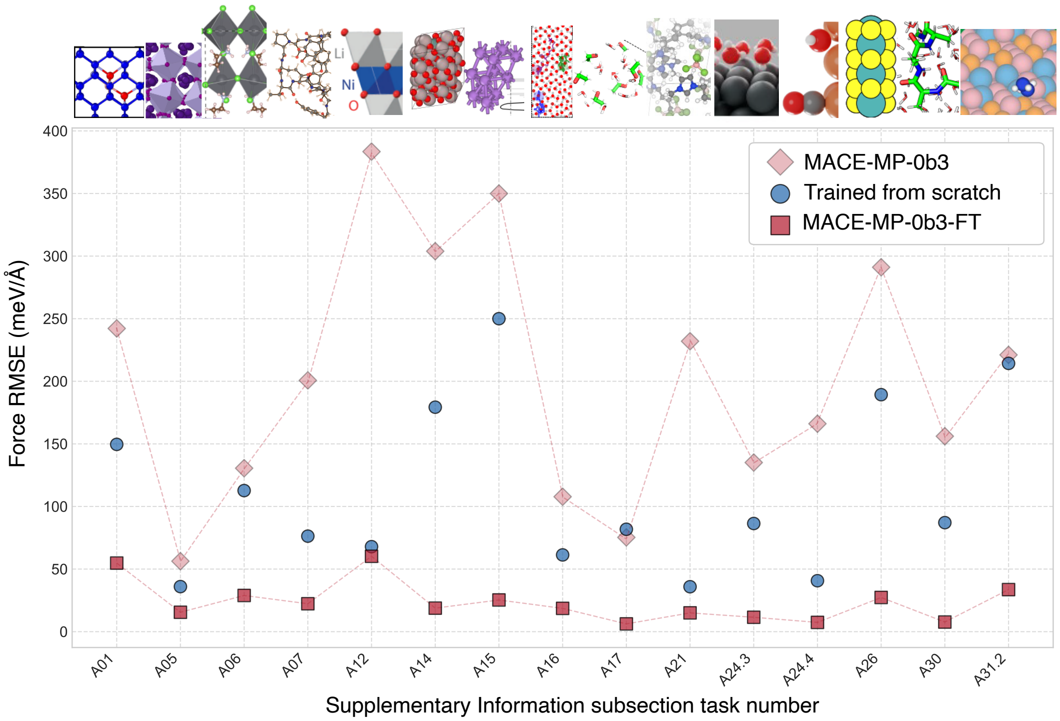

Figure 4: **Fine-tuning.** A comparison of force RMS error on selected applications in the SI for which fine-tuning was performed. The MACE-MP-0b3 model is shown with pink diamonds and the fine-tuned model (MACE-MP-0b3-FT) for each application with red squares. For comparison, in each case we also show the results corresponding to a model trained "from scratch" only to the small amount of fine-tuning data (blue circles).

tograms of energies, forces, stresses, magnetic moments, and element and composition counts, and a discussion of the quantification of the uncertainty in the model predictions.

3 Fine-tuning

Although the multitude of applications demonstrates that MACE-MP-0 is a robust model, it is also clear that, in many cases, it is not accurate enough out of the box to rival or replace *ab initio* calculations. For a selection of examples, we performed fine-tuning on configurations generated using MACE-MP-0, typically via molecular dynamics or other downstream tasks appropriate for the application. We used approximately 100 new configurations for each application during fine-tuning. To prevent catastrophic forgetting (*91*) and retain the robustness of the foundation model, we introduce a new fine-tuning protocol: *multi-head replay fine-tuning*. This approach includes a subset of the foundation model training data in the loss function while fine-tuning on the new data (see Appendix C.2 for details). We train a separate model for each case using this multi-head fine-tuning protocol with replay. Figure 4 shows the resulting force errors, which decrease significantly in every case. For comparison, we also present the force errors of a MACE model trained just on the small fine-tuning dataset. In almost all cases, the force errors of the model trained from scratch are significantly worse than those of the fine-tuned model. In each corresponding subsection of the SI, we demonstrate the performance of the fine-tuned model on application-relevant observables, showing considerable improvement over the original model in every case.

4 Related work: general purpose MLIPs

The development of the MACE-MP-0 models as a foundation models for atomistic materials simulation follows more than a decade of intense activity and progress in making MLIPs for specific materials. General purpose MLIPs – *i.e.,* models that aim to target a wide range of chemical systems spanning many possible combination of elements – are much more recent. Here, we summarise the brief history of such general purpose models as well as the culmination of this trend into the creation of true "foundation models". Within this commentary, we seek to highlight the particular merits of MACE-MP-0 in comparison to existing alternatives. It is worth noting that the reason we call MACE-MP-0 "foundational" is because of how it can be used, as already mentioned in the introduction: the model is suitable for many different applications as a tool for initial exploration, but it likely requires fine-tuning for specific simulation tasks to achieve quantatively accurate predictions.

A key advance towards making general purpose MLIP models was made by MEGNet, introduced in 2019 (*92*). This model, which provides property prediction for inorganic crystals, was trained on minimum energy configurations in the Materials Project (MP) (*93*) that includes most elements of the periodic table (89).Subsequently, models that predict atomic forces were also trained on MP-based datasets, including M3GNet (*94*) and CHGNet (*95*), which were trained on snapshots of DFT relaxations of MP structures, with CHGNet using the MPtrj dataset introduced at the same time (*95*). The ALIGNN-FF model (*96*) was trained on a database of inorganic crystals, JARVIS-DFT (*97*), which covers 89 elements and uses the optB88vdW exchange-correlation functional (*98*). The proprietary GNoME (*99*) (based on the NequIP architecture (*16*)) model also starts from MP, but uses a complex active learning workflow to generate and train on a dataset of inorganic crystals nearly two orders of magnitude larger than MPtrj. The above models were created primarily for the purpose of "materials discovery", i.e. predicting thermodynamic stability of hypothetical inorganic crystals. In addition, they were capable of molecular dynamics for such crystals, and indeed both CHGNet and GNoME were used to study alkali metal ion diffusion in battery materials. More recently, the DPA models (DPA-1 (*100*) and DPA-2 (*101*)) were trained to a wide variety of datasets (with 56 and 73 elements, respectively), a combination of some previously available and some released with the models (altogether 4M configurations). The second paper reports MD results for versions of the baseline model fine-tuned separately to specific systems (e.g. water, solid-state electrolytes, ferroelectric oxide). To date, the most general and transferable force field for molecular dynamics is the PFP model (*102*) (TeaNet architecture (*103*)), also proprietary (including its training set that originally covered 45 elements, recently updated to 72 elements (*104*), and is significantly larger than MP and also covers molecules and surfaces). PFP was demonstrated for running simulations on solid-state ionic conductors, and a molecular adsorption and a heterogeneous catalysis example—systems that formed part of its training data set. There are also ML force fields specialized for organic molecules (with a much more limited number of elements) such as the ANI (and later AimNET) series of models (*105–107*) and the MACE-OFF models (*108*), as well as for metal alloys (*109*). However, there has yet to be a comprehensive demonstration that a single ML potential can describe solid, liquid, and gaseous systems of materials and molecules across the periodic table and well beyond the distribution of the underlying training set. [1]

5 Outlook

The stable MD propagation for a wide range of materials across the periodic table and the DFT-quality simulation (in some cases after fine-tuning) that we have shown here are landmark achievements for a single machine-learned interatomic potential. In this sense, we expect that the present study will have implications for the wider development of the field, beyond any specific model parameterisation. Yet there are a number of limitations of the current ("b3" and "MPA-0") versions of the MACE-MP-0 foundation model.

[1] Since the first preprint version of this manuscript, a number of models have been fitted to the same MPtrj data set and also to larger extended datasets including the Alexandria (*110*) and OMat24. (*111*). Notable models (reported in preprint form) that showed high in-domain accuracy include SevenNet (*112*) (based on the NEquIP architecture (*16*)), GRACE (*113*), Orb (*114, 115*), EquiformerV2 (*111*), MatterSim (*116*)and eSEN (*117*). Of these, the MatterSim models, have been tested in the molecular dynamics for some materials including polymers and surfaces. The GRACE theoretical framework formally generalises MACE, but actual released GRACE models remain in close correspondence with the MACE design choices. To compare with these newer models we also include results for a new model termed MACE-MPA-0 in the SI, which has been trained on an extended dataset including MPtrj and Alexandria.

The exchange–correlation functional used in the MPtrj dataset is PBE (*118*), which must be augmented with Hubbard U terms to improve electronic correlations for particular element combinations (introducing inconsistencies in the PES that must be compensated (*6*)), and dispersion corrections, such as D3 (*29*). Recent developments in DFT are beginning to supersede conventional GGA functionals by achieving improved accuracy at comparable computational cost (*119, 120*), and methods beyond DFT such as hybrid functionals (*121*) and the random phase approximation (*122*) improve upon this even further, but at much larger computational cost. Refitting or fine-tuning the model to a more modern functional is expected to increase its predictive power, and will reduce the need for system-dependent corrections such as the use of Hubbard U terms and dispersion. (Note that the above mentioned inconsistency is not present in the more recent MATPES dataset, (*123*) which removes the Hubbard U correction altogether.)

The MACE architecture that we used to fit the data presently does not contain explicit long-range interactions (beyond the 12 Å receptive field afforded by two steps of message passing), nor does it take into account magnetic or spin degrees of freedom. Despite the success in describing many different chemistries demonstrated herein, there will be observables, particularly in the context of dilute solutions and at interfaces, that cannot be calculated with a short-range model. There are several approaches to incorporating explicit electrostatic interactions into atomistic ML models in the literature (*124–127*), as well as spin degrees of freedom (*95, 128, 129*). In the future, foundation models could undoubtedly benefit from such an extension.

Considering the results for the diverse systems shown in the SI, a particular area where the model clearly needs improvement is describing intermolecular interactions. While the overarching goal of MD stability is achieved, for many systems there is room for improvement in a quantitative sense, for example in obtaining more accurate densities of molecular liquids, such as ethanol-water mixtures (section appendix A.17). The present version of the potential includes a repulsive pair potential (*130*) that helps describe the repulsive interaction of atoms at close range, the accuracy of the model (e.g., in predicting the equation of state) at high pressures is limited due to the absence of data in this regime. This can easily be remedied either by active learning (*116*) or a more systematic approach, e.g. replicating part of the MP dataset at lower and higher densities.

Although we have described an example of a model with wide generalisation, we expect that there will be considerable improvements possible both in the model architecture and in optimising the way in which data is assembled, and the model is fine-tuned. (*101, 131, 132*) It is an open question whether the biggest gains will be obtained by improving the underlying data (both the amount and the consistency) or by scaling the size and expressivity of the model. There is good evidence that reaching higher levels of electronic structure theory (such as improved XC functionals) from a DFT baseline and beyond requires significantly less data than fitting to DFT itself (*106, 133, 134*), and we show an example of this in the SI, where we fine-tune the model to data computed with the r2SCAN functional (*135*).

Finally, there is the tantalising possibility that with some improvements, it will be possible to make an ML force field model that achieves quantitative agreement with explicit electronic structure theory across the full range of chemistry and structure. If this turns out to be true, future foundation models may truly provide a universal model for carrying out atomistic simulations at scale.

6 Methods

MACE All models trained in the paper use the MACE (*19*) architecture implemented in PyTorch (*136*) and employing the *e3nn* library (*137*). The MACE training and evaluation codes are distributed via GitHub under the MIT license, available at `https://github.com/ACEsuit/mace/`. The models used in this paper are available at `https://github.com/ACEsuit/mace-mp/`. MACE is an equivariant message-passing graph tensor network where each layer encodes many-body information of atomic geometry. At each layer, many-body messages are formed using a linear combination of a tensor product basis (*23, 24*). This is constructed by taking tensor products of a sum of two-body permutation-invariant polynomials, expanded in a spherical basis. The final output is the energy contribution of each atom to the total potential energy. For a more detailed description of the architecture, see Refs. (*19*) and (*138*).

Model versions Different model versions have been released based on this work, including the models used in the first version of the manuscript, now named MACE-MP-0a, and the model used in the present version, named MACE-MP-0b3. All previous models can be found at `https://github.com/ACEsuit/mace-mp/`. Unless otherwise stated in the text, all models used in this paper correspond to MACE-MP-0b3. We use the label "MACE-MP-0" to refer to this model series generally.

Hyper-parameters The model referred to in this work uses two MACE layers, a spherical expansion of up to $l_{\max} = 3$, and 4-body messages in each layer (correlation order 3). The model uses a 128-channel dimension for tensor decomposition. We use a radial cutoff of 6 Å and expand the interatomic distances into 10 Bessel functions multiplied by a smooth polynomial cutoff function to construct radial features, in turn fed into a fully-connected feed-forward neural network with three hidden layers of 64 hidden units and SiLU non-linearities. We fit an $L = 1$ model, corresponding to a "medium sized" model, as it represents a good compromise. More efficient models that only pass invariants during the message passing step ($L = 0$) or those that pass higher order tensors ($L \geq 2$) are straightforward to train, and can form part of the accuracy/efficiency tradeoff in selecting the optimal model in the future. The irreducible representations of the messages have alternating parity (in *e3nn* notation, `128x0e + 128x1o`).

Distance transforms and pair repulsion Smooth behavior of the potential at close approach is essential for a broadly applicable model. We use a combination of Ziegler–Biersack–Littmark (ZBL) (*139*) core potential to the short-range repulsive forces, and distance transformation to smoothly connect this behavior to equilibrium interactions. The ZBL energy is given by,

$$E_{\mathrm{ZBL}} = \sum_j \frac{14.3996 \cdot Z_i \cdot Z_j}{r_{ij}} \cdot \phi(r_{ij}/a) \cdot \mathrm{Envelope}(r_{ij}, r_{\max}, p), \tag{1}$$

$$\phi(r/a) = c_0 e^{-3.2(r/a)} + c_1 e^{-0.9423(r/a)} + c_2 e^{-0.4028(r/a)} + c_3 e^{-0.2016(r/a)}. \tag{2}$$

where Z_u and Z_v are the atomic numbers of the interacting atoms, and r_{ij} is the interatomic distance between atoms i and j. The screening length a is given by $a = 0.529 \cdot a_{\mathrm{prefactor}}/(Z_i^{a_{\exp}} + Z_j^{a_{\exp}})$. The coefficients are $c = \{0.1818, 0.5099, 0.2802, 0.02817\}$. The maximum cutoff radius is defined as $r_{\max} = R_{\mathrm{cov}}(Z_u) + R_{\mathrm{cov}}(Z_v)$. The envelope function $\mathrm{Envelope}(r, r_{\max}, p)$ is a polynomial cutoff function applied to smooth the potential. We use the same envelope as the radial basis. To smoothly transition from the the ZBL to the MACE energy, we use the Agnesi distance transform (*140*),

$$y_{ij} = \left(1 + \frac{a \cdot (r_{ij}/r_0)^q}{1 + (r_{ij}/r_0)^{q-p}}\right)^{-1}. \tag{3}$$

where y_{ij} is the transformed distance, and the parameters a, q, and p control the shape of the transformation, $r_0 = \frac{1}{2}(R_{\mathrm{cov}}(Z_u) + R_{\mathrm{cov}}(Z_v))$. We then evaluate the radial basis in this transformed space instead of directly on the distances.

Normalization To ensure internal normalization of the weights and smooth extrapolation to high pressure systems, we divide the atomic basis in each layer by a learnable quantity called density normalization e_i,

$$e_i = 1 + \sum_j \tanh\left(\mathrm{SiLU}\big(\big[\sum_k W_k B_k(r_{ij})\big]\big)^2\right) \tag{4}$$

where B denotes a set of Bessel basis and W are learnable weights. The predicted density normalization varies between 1 and the number of neighbors of atom i, depending on the local environment. This normalization corresponds to a smooth version of the node degree normalization in graph neural networks (*141*). The node energy ϵ_a of atom a is shifted by the isolated atoms energies. Therefore, the prediction of the energy for the whole structure is constructed as

$$\hat{E} = \sum_{a=1}^{N}\left[\sigma\left(\sum_{k=1}^{K}\epsilon_a^{(k)}\right) + \mu_{Z_a}\right]$$

where K denotes the total number of message passing layers and $\epsilon_a^{(k)}$ is the energy of atom a at layer k. μ and σ are the isolated atomic energies and the mean square of the atomic forces computed on the training set. The predicted forces and stresses are computed as derivatives of the total energy with respect to the atomic positions and the strain tensor, respectively.

Training loss The models were trained using a weighted sum of Huber losses of energy, forces, and stress:

$$\begin{aligned}\mathcal{L} = {} & \frac{\lambda_E}{N_b}\sum_{b=1}^{N_b}\mathcal{L}_{\text{Huber}}\left(\frac{\hat{E}_b}{N_a}, \frac{E_b}{N_a}, \delta_E\right) \\ & + \frac{\lambda_F}{3\sum_{b=1}^{N_b} N_a}\sum_{b=1}^{N_b}\sum_{a=1}^{N_a}\sum_{i=1}^{3}\mathcal{L}^{\star}_{\text{Huber}}\left(-\frac{\partial \hat{E}_b}{\partial r_{b,a,i}}, F_{b,a,i}, \delta_F\right) \\ & + \frac{\lambda_\sigma}{9N_b}\sum_{b=1}^{N_b}\sum_{i=1}^{3}\sum_{j=1}^{3}\mathcal{L}_{\text{Huber}}\left(\frac{1}{V_b}\frac{\partial \hat{E}_b}{\partial \varepsilon_{b,ij}}, \sigma_{b,ij}, \delta_\sigma\right),\end{aligned} \tag{5}$$

where $\lambda_E, \lambda_F, \lambda_\sigma$ are predetermined weights of energy (E), forces (F), and stress (σ) losses, the symbols under a hat correspond to predicted values, and N_b and N_a are the batch size and the number of atoms in each structure. In the last term involving the stress, ε_b and σ_b correspond to the strain and stress tensors, respectively. We used $(\lambda_E, \lambda_F, \lambda_\sigma) = (1, 10, 10)$ and Huber deltas of $\delta_E = 0.01, \delta_F = 0.01, \delta_\sigma = 0.01$. We use a conditional Huber loss $\mathcal{L}^{\star}_{\text{Huber}}$ for forces, where the Huber delta δ_F is adaptive to the force magnitude on each atom. The Huber delta δ_F decreases step-wise by a factor from 1.0 to 0.1 as the atomic force increases from 0 to 300 eV/Å. For more details, see the section C.1 in the SI.

Optimization The models are trained with the AMSGrad (*142*) variant of Adam (*143*) with default parameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$. We use a learning rate of 0.001 and a exponential moving average (EMA) learning scheduler with decaying factor of 0.99999. We employ a gradient clipping of 100. The training curves for the medium model is presented in Fig. S63 in the SI. Model is trained for 100 epochs on 40–80 NVIDIA H100 GPUs across 10–20 nodes. Training the medium-sized model took approx. 2,600 GPU hours. We find that MACE-MP-0 achieves an energy MAE of 18 meV/atom and a force MAE of 39 meV/Å for the medium model. After fine-tuning with higher weights for energies for an additional 50 epochs, the small model is able to achieve an energy MAE of 13 meV/atom (see SI C.1).

Performance The speed of evaluation of the MACE-MP-0 model depends on the atomic density, hardware, floating point precision, size of model, *etc.* (see section SI A.32 for details), but a rough guide is that on a single NVIDIA A100 GPU with 80GB of RAM, it can do several nanoseconds per day for 1000 atoms. When run in parallel using domain decomposition, weak scaling at 0.1 ns/day is perfect up to 32,000 atoms and 64 GPUs for a dense metallic alloy.